

Lab 4 security (SCREENSHOTS)

```
PS C:\WINDOWS\system32>
>> mkdir C:\Users\LENOVO\LAB4_WORKSPACE
```

Répertoire : C:\Users\LENOVO

Mode		LastWriteTime	Length	Name
d-----		07/05/2026 15:37		LAB4_WORKSPACE

```
PS C:\WINDOWS\system32>
```

```
PS C:\Users\LENOVO\LAB4_WORKSPACE> adb devices
List of devices attached
emulator-5554    device
```

```
PS C:\Users\LENOVO\LAB4_WORKSPACE>
```

```
PS C:\WINDOWS\system32>
>> adb shell pm path owasp.mstg.uncrackable1
package:/data/app/~~UMfm8hyXnHsK5ux7zBSP4A==/owasp.mstg.uncrackable1-gH48Jl5o
wLYsWH-02zRaIQ==/base.apk
PS C:\WINDOWS\system32>
```

```
PS C:\Users\LENOVO\LAB4_WORKSPACE>
>> cd C:\Tools
PS C:\Tools> dir
```

Répertoire : C:\Tools

Mode		LastWriteTime	Length	Name
d-----		26/04/2026 15:55		dex2jar
d-----		26/04/2026 16:33		jadx
d-----		26/04/2026 15:56		jd-gui

```
PS C:\Tools>
```

```
PS C:\Tools> cd C:\Users\LENOVO\LAB4_WORKSPACE
PS C:\Users\LENOVO\LAB4_WORKSPACE> adb shell pm path owasp.mstg.uncrackable1
package:/data/app/~~UMfm8hyXnHsK5ux7zBSP4A==/owasp.mstg.uncrackable1-gH48Jl5o
wLYsWH-02zRaIQ==/base.apk
PS C:\Users\LENOVO\LAB4_WORKSPACE> adb pull "/data/app/~~UMfm8hyXnHsK5ux7zBSP
4A==/owasp.mstg.uncrackable1-gH48Jl5owLYsWH-02zRaIQ==/base.apk" uncrackable1.
apk
/data/app/~~UMfm8hyXnHsK5ux7zBSP4A==/...ped. 4.6 MB/s (66651 bytes in 0.014s)
PS C:\Users\LENOVO\LAB4_WORKSPACE>
```

```
PS C:\Users\LENOVO\LAB4_WORKSPACE> ls uncrackable1.apk
```

Répertoire : C:\Users\LENOVO\LAB4_WORKSPACE

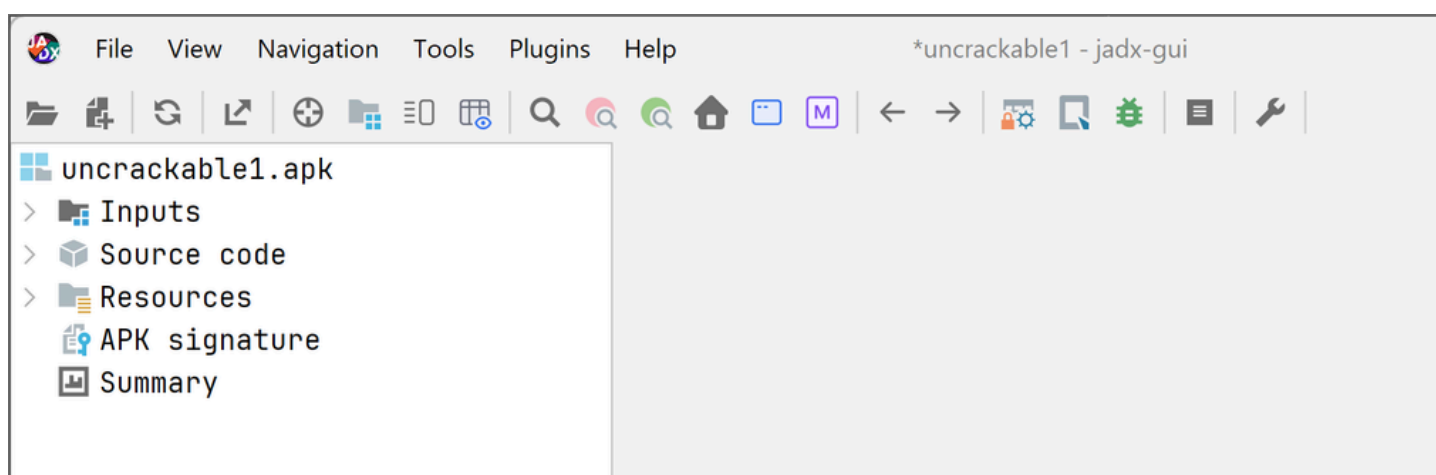
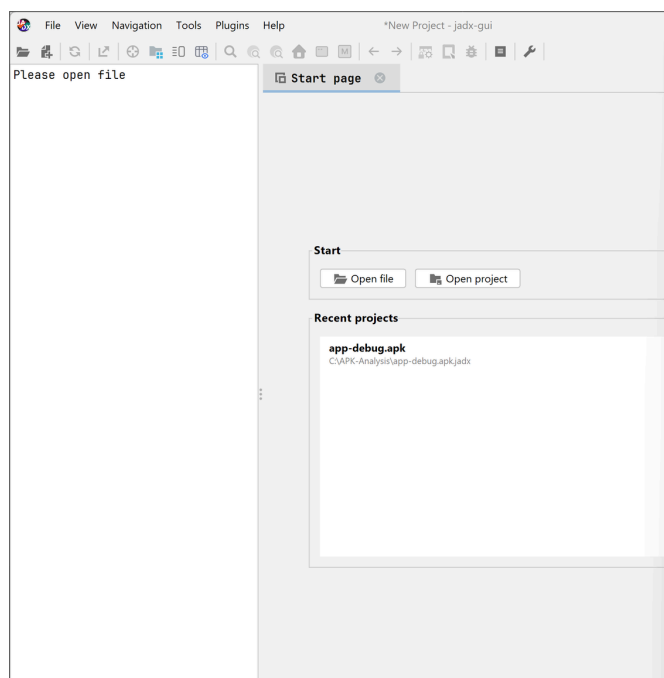
Mode	LastWriteTime	Length	Name
----	-----	-----	----
-a----	07/05/2026 15:49	66651	uncrackable1.apk

```
PS C:\Users\LENOVO\LAB4_WORKSPACE>
```

Administrateur : Windows PowerShell

```
PS C:\Tools\jadx> C:\Tools\jadx\jadx-gui-1.5.5.exe
```

```
PS C:\Tools\jadx>
```



File View Navigation Tools Plugins Help *uncrackable1 - jadx-gui

uncrackable1.apk

- Inputs
 - Files
- Source code
 - owasp.mstg.uncrackable1
 - R
 - sg.vantagepoint
- Resources
 - APK signature
 - Summary

Summary x R x

```
1 package owasp.mstg.uncrackable1;
2
3 /* JADX INFO: This class is generated by JADX */
4 public final class R {
5
6     public static final class dimen {
7         public static final int activity_horizontal_margin = 0x7f010000;
8         public static final int activity_vertical_margin = 0x7f010001;
9     }
10
11     public static final class id {
12         public static final int action_settings = 0x7f020000;
13         public static final int edit_text = 0x7f020001;
14     }
15
16     public static final class layout {
17         public static final int activity_main = 0x7f030000;
18     }
19
20     public static final class menu {
21         public static final int menu_main = 0x7f040000;
22     }
23
24     public static final class mipmap {
25         public static final int ic_launcher = 0x7f050000;
26     }
27
28     public static final class string {
29         public static final int action_settings = 0x7f060000;
30         public static final int app_name = 0x7f060001;
31         public static final int button_verify = 0x7f060002;
32         public static final int edit_text = 0x7f060003;
33         public static final int thanks = 0x7f060004;
34     }
35
36     public static final class style {
37         public static final int AppTheme = 0x7f070000;
38     }
39 }
```

Code Smali Simple Fallback ☐ Split view

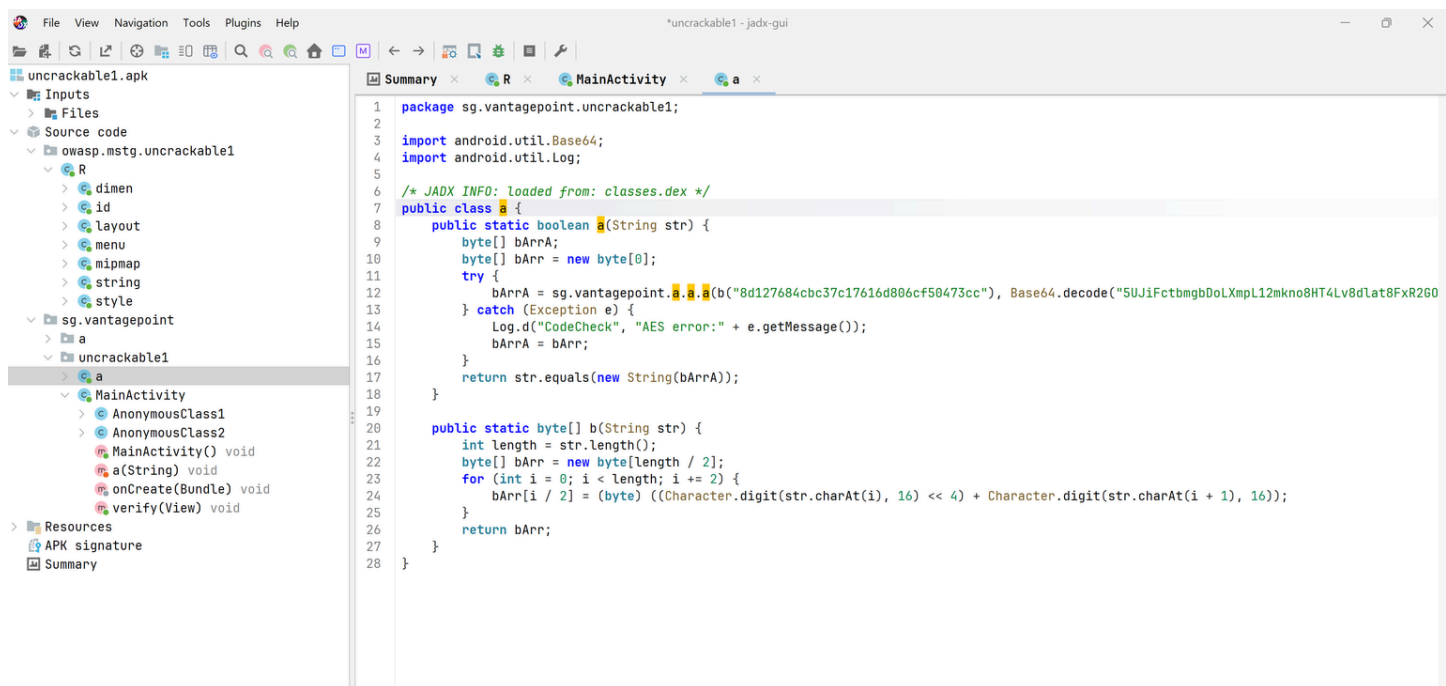
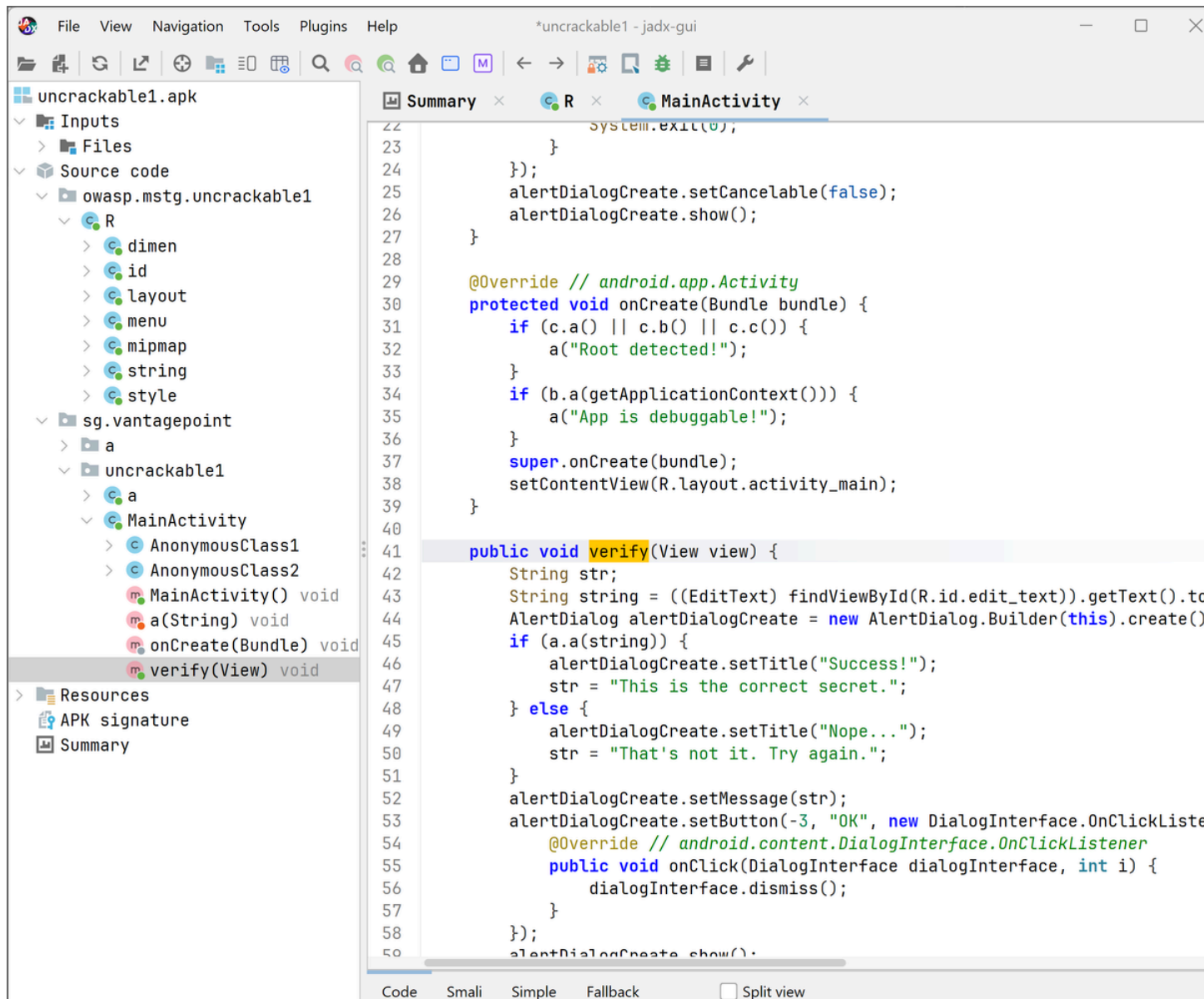
File View Navigation Tools Plugins Help *uncrackable1 - jadx-gui

uncrackable1.apk

- Inputs
 - Files
- Source code
 - owasp.mstg.uncrackable1
 - R
 - dimen
 - id
 - layout
 - menu
 - mipmap
 - string
 - style
 - sg.vantagepoint
 - a
 - uncrackable1
 - a
 - MainActivity
 - AnonymousClass1
 - AnonymousClass2
 - MainActivity() void
 - a(String) void
 - onCreate(Bundle) void
 - verify(View) void
- Resources
 - APK signature
 - Summary

Summary x R x MainActivity x

```
1 package sg.vantagepoint.uncrackable1;
2
3 import android.app.Activity;
4 import android.app.AlertDialog;
5 import android.content.DialogInterface;
6 import android.os.Bundle;
7 import android.view.View;
8 import android.widget.EditText;
9 import owasp.mstg.uncrackable1.R;
10 import sg.vantagepoint.a.b;
11 import sg.vantagepoint.a.c;
12
13 /* JADX INFO: loaded from: classes.dex */
14 public class MainActivity extends Activity {
15     private void a(String str) {
16         AlertDialog alertDialogCreate = new AlertDialog.Builder(this).create()
17         alertDialogCreate.setTitle(str);
18         alertDialogCreate.setMessage("This is unacceptable. The app is now goin
19         alertDialogCreate.setButton(-3, "OK", new DialogInterface.OnClickListener
20             @Override // android.content.DialogInterface.OnClickListener
21             public void onClick(DialogInterface dialogInterface, int i) {
22                 System.exit(0);
23             }
24         });
25         alertDialogCreate.setCancelable(false);
26         alertDialogCreate.show();
27     }
28
29     @Override // android.app.Activity
30     protected void onCreate(Bundle bundle) {
31         if (c.a() || c.b() || c.c()) {
32             a("Root detected!");
33         }
34         if (b.a(getApplicationContext())) {
35             a("App is debuggable!");
36         }
37         super.onCreate(bundle);
```




```

/* JADX INFO: loaded from: classes.dex */
public class a {
    public static boolean a(String str) {
        byte[] bArrA;
        byte[] bArr = new byte[0];
        try {
            bArrA = sg.vantagepoint.a.a.a(b("8d127684cbc37c17616d806cf50473cc"), Base64.decode("
        } catch (Exception e) {
            Log.d("CodeCheck", "AES error:" + e.getMessage());
            bArrA = bArr;
        }
    }
}

```

The screenshot shows the JADX GUI interface. The left sidebar displays the project structure for 'uncrackable1.apk', including 'Inputs', 'Files', 'Source code', and 'sg.vantagepoint'. The main editor window shows the 'MainActivity' class. The search bar at the top right contains the text 'This is the correct secret', and the results pane shows 1 result. The code in the editor is as follows:

```

20      alertDialogCreate.show(),
21  }
22
23  @Override // android.app.Activity
24  protected void onCreate(Bundle bundle) {
25      if (c.a() || c.b() || c.c()) {
26          a("Root detected!");
27      }
28      if (b.a(getApplicationContext())) {
29          a("App is debuggable!");
30      }
31      super.onCreate(bundle);
32      setContentView(R.layout.activity_main);
33  }
34
35  public void verify(View view) {
36      String str;
37      String string = ((EditText) findViewById(R.id.edit_text)).getText().toString();
38      AlertDialog alertDialogCreate = new AlertDialog.Builder(this).create();
39      if (a.a(string)) {
40          alertDialogCreate.setTitle("Success!");
41          str = "This is the correct secret.";
42      } else {
43          alertDialogCreate.setTitle("Nope...");
44          str = "That's not it. Try again.";
45      }
46      alertDialogCreate.setMessage(str);
47      alertDialogCreate.setButton(-3, "OK", new DialogInterface.OnClickListener() { // from class
48          @Override // android.content.DialogInterface.OnClickListener
49          public void onClick(DialogInterface dialogInterface, int i) {
50              dialogInterface.dismiss();
51          }
52      });
53      alertDialogCreate.show();
54  }
55
56  }
57
58  }
59

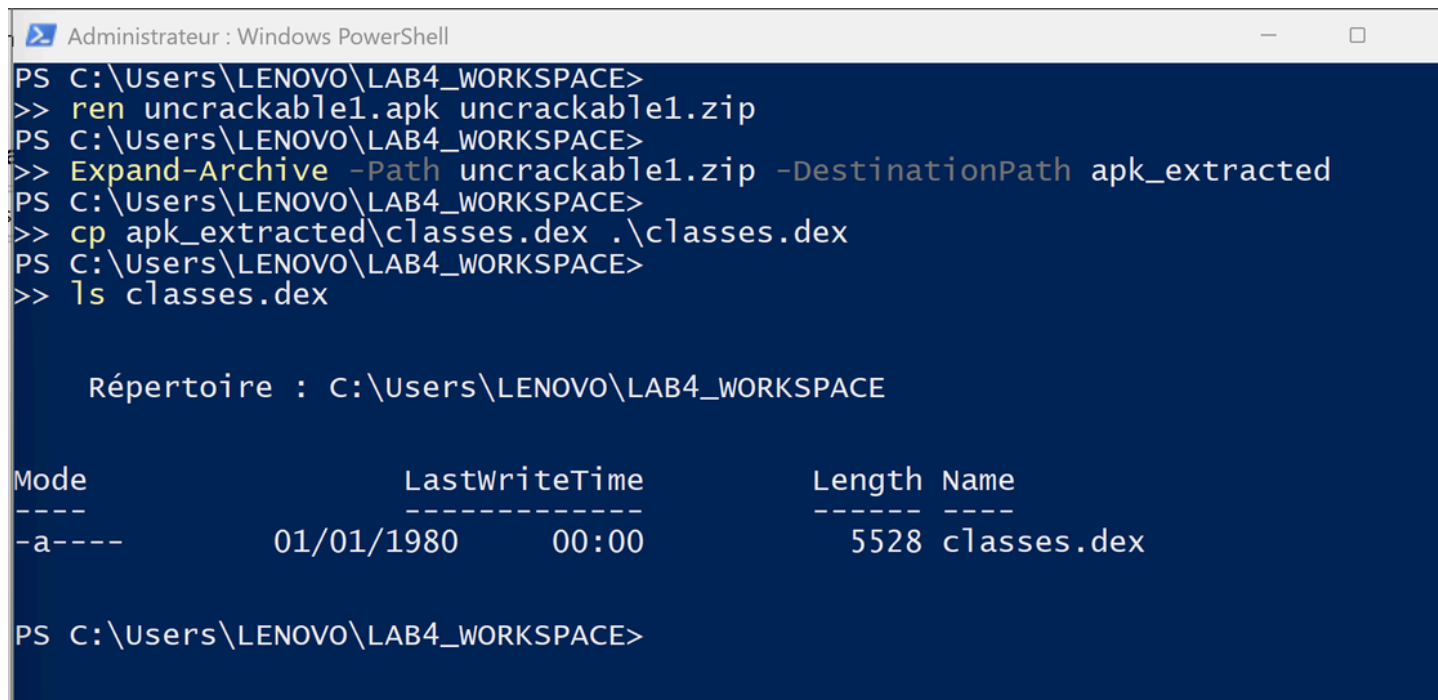
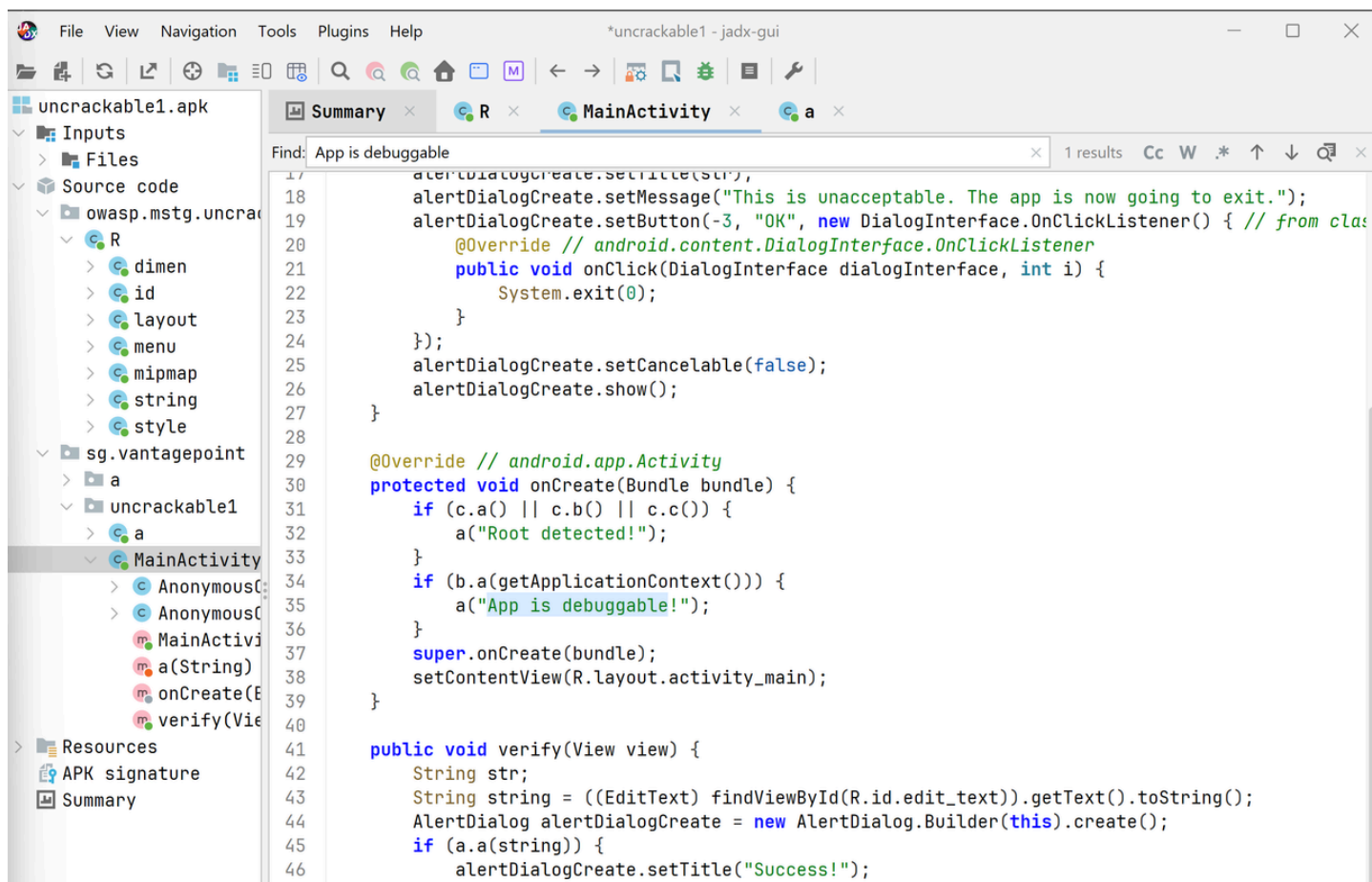
```

The screenshot shows the JADX GUI interface. The left sidebar displays the project structure for 'uncrackable1.apk', including 'Inputs', 'Files', 'Source code', and 'sg.vantagepoint'. The main editor window shows the 'MainActivity' class. The search bar at the top right contains the text 'Root detected', and the results pane shows 1 result. The code in the editor is as follows:

```

17      alertDialogCreate.setTitle(str);
18      alertDialogCreate.setMessage("This is unacceptable. The app is now going to exit.");
19      alertDialogCreate.setButton(-3, "OK", new DialogInterface.OnClickListener() { // from class
20          @Override // android.content.DialogInterface.OnClickListener
21          public void onClick(DialogInterface dialogInterface, int i) {
22              System.exit(0);
23          }
24      });
25      alertDialogCreate.setCancelable(false);
26      alertDialogCreate.show();
27  }
28
29  @Override // android.app.Activity
30  protected void onCreate(Bundle bundle) {
31      if (c.a() || c.b() || c.c()) {
32          a("Root detected!");
33      }
34  }
35

```



```
Administrateur : Windows PowerShell
PS C:\Users\LENOVO\LAB4_WORKSPACE> dir C:\Tools\dex2jar

Répertoire : C:\Tools\dex2jar

Mode                LastWriteTime         Length Name
----                -
d-----          26/04/2026      15:55         lib
-a-----          26/04/2026      15:55      80384 jadx-gui-1.5.5.exe

PS C:\Users\LENOVO\LAB4_WORKSPACE>
```

```
PS C:\Tools> dir C:\Tools\dex2jar

Répertoire : C:\Tools\dex2jar

Mode                LastWriteTime         Length Name
----                -
d-----          07/05/2026      16:17         dex-tools-2.4.36

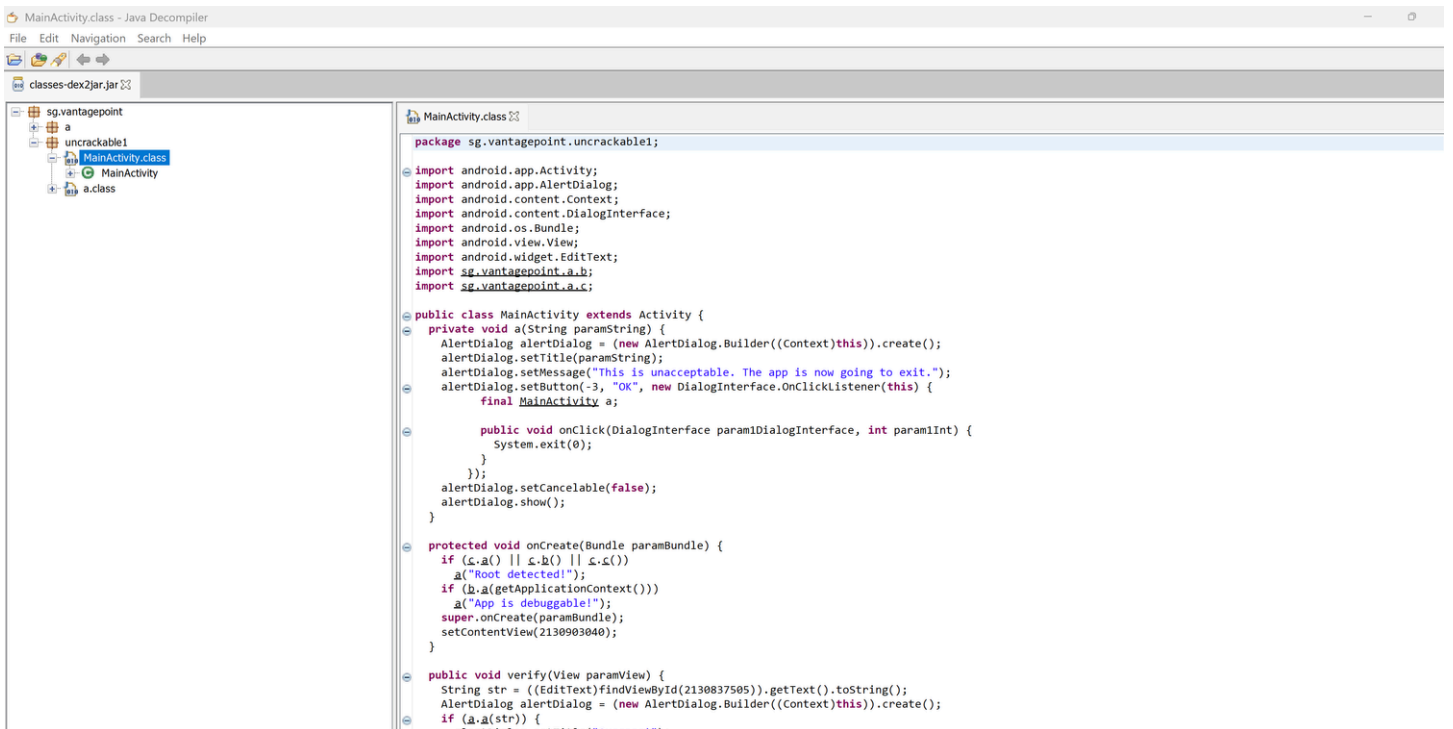
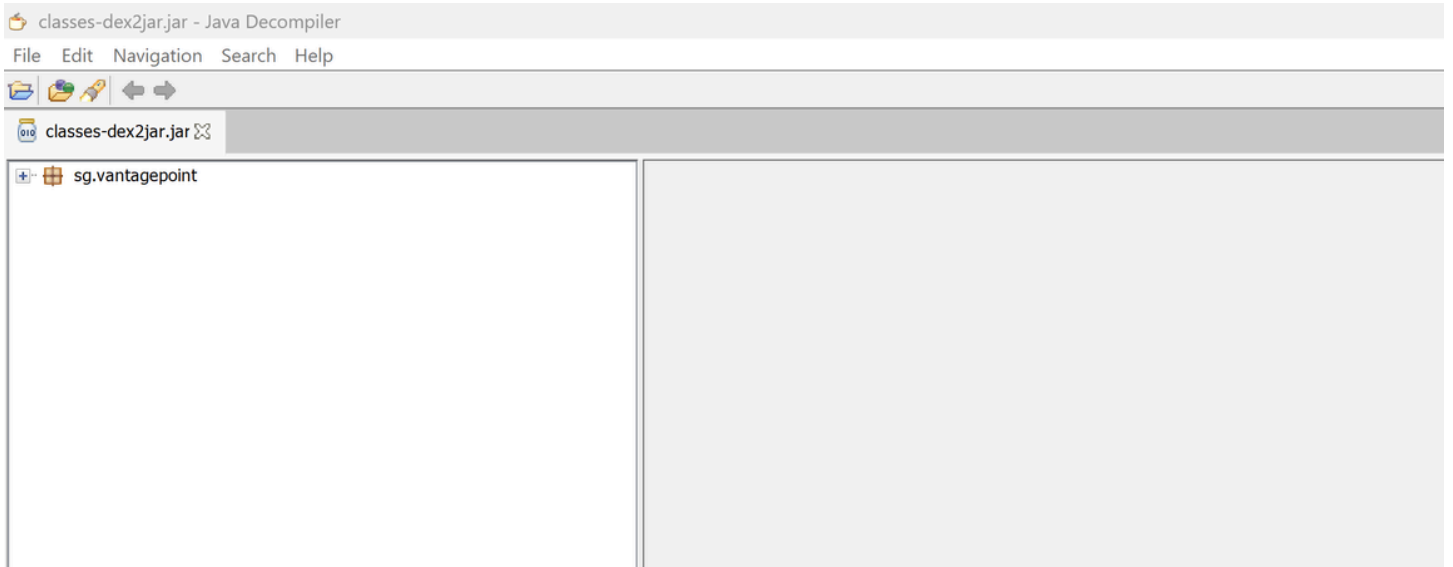
PS C:\Tools>
```

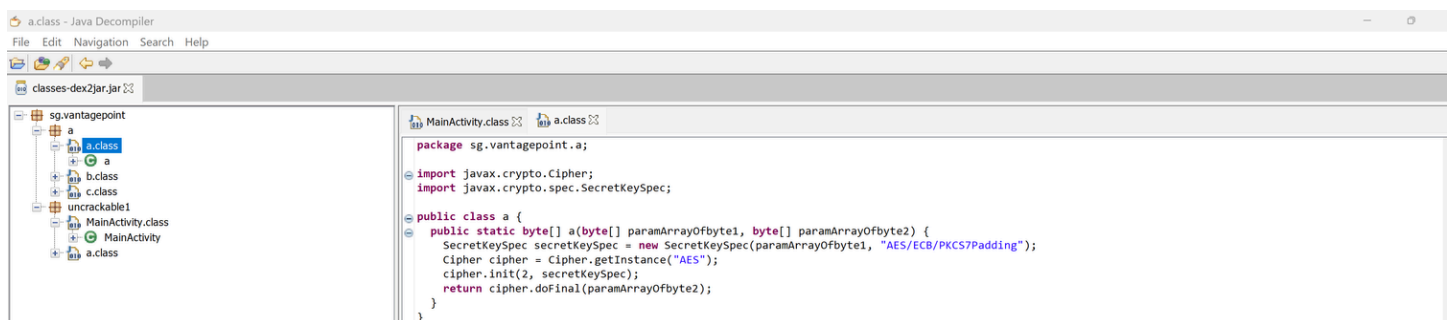
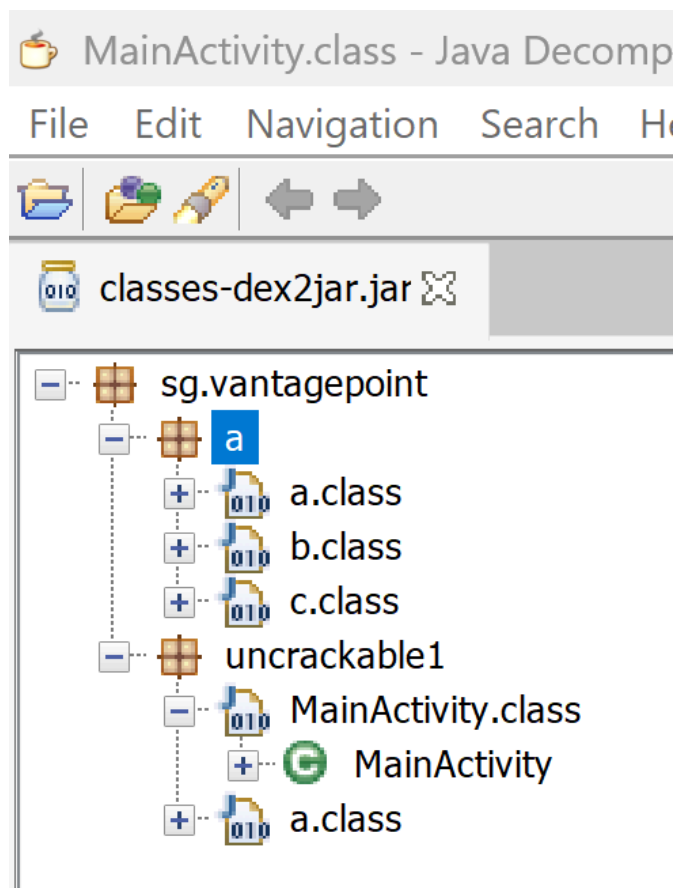
```
PS C:\Tools>
PS C:\Tools>
>> cd C:\Users\LENOVO\LAB4_WORKSPACE
PS C:\Users\LENOVO\LAB4_WORKSPACE>
>> C:\Tools\dex2jar\dex-tools-2.4.36\d2j-dex2jar.bat classes.dex
dex2jar classes.dex -> .\classes-dex2jar.jar
PS C:\Users\LENOVO\LAB4_WORKSPACE>
```

```
Répertoire : C:\Tools\jd-gui\jd-gui-windows-1.6.6

Mode                LastWriteTime         Length Name
----                -
-a-----          07/05/2026      16:27     1608662 jd-gui.exe
-a-----          07/05/2026      16:27       35836 LICENSE
-a-----          07/05/2026      16:27        309 NOTICE
-a-----          07/05/2026      16:27       2546 README.md

PS C:\Users\LENOVO\LAB4_WORKSPACE> C:\Tools\jd-gui\jd-gui-windows-1.6.6\jd-gu
i.exe
PS C:\Users\LENOVO\LAB4_WORKSPACE>
```



```
PS C:\WINDOWS\system32>
>> cd C:\Users\LENOVO\LAB4_WORKSPACE
PS C:\Users\LENOVO\LAB4_WORKSPACE>
>> rm classes.dex
>> cd C:\Users\LENOVO\LAB4_WORKSPACE
PS C:\Users\LENOVO\LAB4_WORKSPACE>
>> rm classes-dex2jar.jar
>> rm classes-dex2jar.jar
```

```
PS C:\Users\LENOVO\LAB4_WORKSPACE>
>> rm uncrackable1.zip
PS C:\Users\LENOVO\LAB4_WORKSPACE>
>> rm -r apk_extracted
PS C:\Users\LENOVO\LAB4_WORKSPACE>
```